

3.8 Errors and Exceptions (Try-Except)

ایکسپشن (Exception) چلتے پروگرام کی غلطی ہے، جیسے غیر موجود فائل کھولنا۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 16 of 17

3.8 Errors and Exceptions (Try-Except)

LEARN IT

An **exception** is an error raised while the program is running — for example opening a file that is not there. Without handling, Python prints a traceback and **stops**. A `try / except` block catches the error and lets you show a clear message instead of crashing.

Error type	Meaning
<code>FileNotFoundError</code>	The path does not exist.
<code>PermissionError</code>	The operating system will not let you read or write that file.
<code>IOError</code>	A general input/output problem (disk full, disk disconnected).

Put risky code in `try`. Put the recovery in `except ErrorType`. A final `except Exception as e` can catch anything else, using `e` as the error object.

```
try:
    with open('missing_data.txt', 'r') as f:
        content = f.read()
        print(content)
except FileNotFoundError:
    print("Error: The file does not exist. Please check the name.")
except Exception as e:
    print(f"An unexpected error occurred: {e}")
```

OUTPUT

```
Error: The file does not exist. Please check the name.
```

Without `try`, the same open raises:

OUTPUT

```
Traceback (most recent call last):
  File "missing.py", line 1, in <module>
    open('missing_data.txt', 'r')
FileNotFoundError: [Errno 2] No such file or directory: 'missing_data.txt'
```

WORKED EXAMPLE

When the file exists, the except blocks are skipped.

```
with open('ok.txt', 'w') as f:
    f.write("Present\n")

try:
    with open('ok.txt', 'r') as f:
        content = f.read()
        print(content, end="")
except FileNotFoundError:
    print("Error: The file does not exist. Please check the name.")
```

OUTPUT

Present

Professional software does not dump a red traceback on a user. If a file is missing or a disk fails, the user should see a calm sentence such as "Please check the file name" or "Please try again later."

REMEMBER IT

try = attempt; except = backup plan. Catch the **specific** error first (`FileNotFoundError`), then a general `Exception` if you need it. Order matters: a bare `except Exception` first would swallow the specific error.

COMMON MISTAKE

Do not wrap your entire program in one silent `except:` that ignores every mistake. You would hide real bugs. Catch the errors you expect, and keep the message honest. `'w'` and `'a'` create a missing file, so they do not raise `FileNotFoundError`; `'r'` does.

اردو میں سمجھیں

ایکسپشن (Exception) چلتے پروگرام کی غلطی ہے، جیسے غیر موجود فائل کھولنا۔ try میں خطرناک کوڈ لکھیں اور `except FileNotFoundError` میں صاف پیغام دکھائیں تاکہ پروگرام کریش نہ ہو۔ پیشہ ورانہ سافٹ ویئر صارف کو سرخ ٹریس بیک کی بجائے سمجھ آنے والا جملہ دکھاتا ہے۔

CHECK YOURSELF

1. Which exception is raised when `open('missing_data.txt', 'r')` cannot find the file?
2. Does the sample program crash, or print a message, when the file is missing?

Answers: 1. `FileNotFoundError`. 2. It prints "Error: The file does not exist. Please check the name." and continues.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. try/except is for:

- a. Handling runtime errors without crashing the whole program
- b. Fixing syntax errors (missing colons)
- c. Faster Big O
- d. Drawing pie charts

MCQ 2. FileNotFoundError is raised when:

- a. open r on a missing path
- b. 1/0
- c. int("a")
- d. A logic bug that prints 3 instead of 4

MCQ 3. except ValueError catches:

- a. int("hi")
- b. Missing files
- c. SyntaxError of a colon
- d. Logic silent bugs

MCQ 4. except Exception is:

- a. Broad — use with care
- b. The way to hide all bugs always
- c. A sort
- d. A key

MCQ 5. else on try runs when:

- a. No exception occurred
- b. Always
- c. Only on errors
- d. On syntax errors

MCQ 6. finally runs:

- a. Always (cleanup)
- b. Only on success
- c. Only on error
- d. Never with with

Extra short questions

1. Why exceptions in file programs?
2. Show a try around int(input).
3. Does try fix logic errors?
4. Order of except clauses.

Extra long question

1. Write a reader that tries to open marks.txt, handles missing file, skips bad lines with ValueError, and prints a count of good rows.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Missing files and bad lines are expected; crash is a poor UX.
2. try: `n=int(input())` except `ValueError`: `print("Enter a number")`.
3. No — wrong formula still wrong.
4. Specific first, broader later.

Long

1. Full code. Sample: file missing message; file with 80, x, 90 → 2 good. Mention not using bare except.

Next lecture: 3.9 Mini File Handling Project: Grade Tracker